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UKUDLA Data Science Certificate

Data Preparation

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UKUDLA

African German Centre

for Sustainable and Resilient Food
Systems and Applied Agricultural
and Food Data Science



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Integrated data are not automatically analysis-ready

The sources are managed and joined

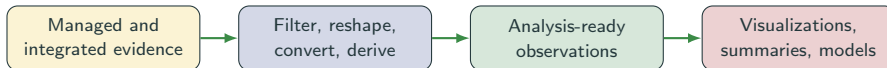
- inputs are identified and documented;
- keys and coverage are validated;
- identifiers, space and time are aligned; and
- the join is audited.

The analysis still needs decisions

- which observations are retained?
- which shape should the table have?
- which units and types are required?
- which variables should be derived?

Data preparation turns managed evidence into a documented representation suited to a defined analysis.

Preparation choices affect the result



- Dropping a record changes the population represented.
- Aggregation hides variation and requires a summary rule.
- Unit conversion changes values but should preserve meaning.
- Derived variables encode assumptions that later results inherit.

This session moves from **motivation**, through the essential **concepts**, to a reproducible **application** for an analytical dataset.

Related activities answer different questions

Activity	Central question	Typical evidence
Validation	Do data meet documented expectations?	Checks and findings
Cleaning	Is a value demonstrably erroneous, and how is it corrected?	Justified correction record
Integration	How are compatible observations connected across sources?	Join contract and audit
Preparation	How are data represented for the intended analysis?	Reproducible transformations

A surprising value is not automatically an error. Preserve the evidence, investigate its meaning, and separate correction from transformation.

Begin with the target observation

Analysis contract

Before transforming, state:

- intended population;
- observation grain;
- candidate key;
- outcome and other variables;
- units and time periods; and
- missing-value policy.

Example analysis target

One row represents one region and year, containing an outcome and explanatory measures aligned to that period.

Candidate key

`region_id + year`

The target grain determines which transformations are valid.

Reshape structure without hiding meaning

Long source representation

Unit	Measure	Value
Plot 12	Biomass	2.34
Plot 12	Soil moisture	18.6
Plot 12	Nitrogen	40

Wide analytical representation

Unit	Biomass	Soil moisture
Plot 12	2.34	18.6

One target unit becomes one row with a separate variable for each retained measure.

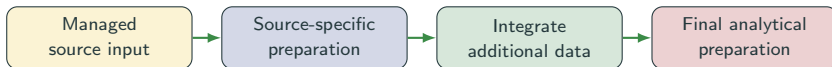
Reshaping is valid only after the source key is known and every output cell is expected to receive at most one value.

Make every transformation explicit

Operation	Question	Evidence to preserve
Select/filter	Which population and variables remain?	Inclusion rule and row counts
Parse/recode	How are types, dates, labels or codes interpreted?	Mapping and failed parses
Convert	Are definitions and dimensions compatible?	Formula, source and target units
Reshape	What becomes an observation or variable?	Input and output grain and keys
Handle missingness	Is absence unknown, structural or introduced?	Missingness before and after
Derive	What assumption defines the new variable?	Formula, inputs and valid domain

Do not silently delete duplicates, replace missing values, cap outliers, or coerce failed values merely to make code complete.

Preparation can occur around integration



- Source-specific preparation makes source grains compatible.
- Integration connects compatible observations across sources.
- Final preparation selects or derives variables for a stated analysis.
- Each stage writes a new artifact; no stage edits a managed input.

Workflow order is determined by dependencies, not by topic boundaries.

Prepare an analysis-specific dataset

Validate before reshaping

- expected variable–unit pairs;
- unique unit–occasion–measure keys;
- known unit, treatment and period coverage; and
- numeric values and missingness.

Transform deliberately

- retain observations relevant to the question;
- reshape measures into analytical variables;
- harmonize units under documented formulas;
- derive transformations only on valid domains; and
- arrange one row per target observation.

The conversion formula, log domain, output grain, row count and candidate key are part of the preparation contract.

Audit preparation and preserve lineage

Preparation audit

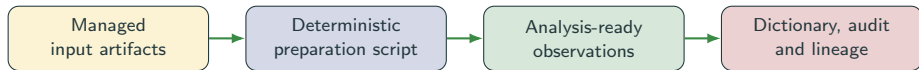
- input and output rows;
- unique input and output keys;
- failed parses or mappings;
- missingness before and after;
- valid units and ranges; and
- unchanged managed inputs.

Lineage record

- input and output artifacts;
- script and relevant parameters;
- filters and reshaping rules;
- conversion and derivation formulas;
- information lost; and
- dictionary updates.

Transformations learned from observed distributions—such as imputation or scaling—must later be estimated from training data only to prevent leakage.

Key message: analysis-ready is purpose-specific



1. State the target population, grain, key, variables and units.
2. Validate each assumption before applying its transformation.
3. Write derived artifacts instead of editing managed inputs.
4. Check row counts, keys, missingness, units and valid domains.
5. Document how every output variable was created.

Analysis-ready means ready for a stated purpose—not universally clean or final.

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